

HW 2 - Convolutional Neural Networks (CNNs)

Google Colab [Optional]

If you are using Google Colab, configure and run the cell below to mount your Google Drive and set up the correct working directory for the assignment. You may ignore this cell if you are working locally.

```
In [ ]: from google.colab import drive
import os

# 1. Mounting Google Drive: This allows Colab to access files in your Google Drive
drive.mount('/content/drive')

# 2. Tell Colab where to find your assignment files and where to save your work

# TODO: Enter the relative path in your Google Drive of the assignment.
# FOLDERNAME = "hw2_student_version" # e.g. 'cs7643/ps0/'
FOLDERNAME = "Deep Learning Homework/hw2"

assert FOLDERNAME is not None, "[!] Enter the foldername."

FULL_PATH = "/content/drive/MyDrive/" + FOLDERNAME
assert os.path.exists(FULL_PATH), "Make sure your FOLDERNAME is correct"
```

```
In [ ]: %cd $FULL_PATH
```

```
In [ ]: !ls "/content/drive/MyDrive/Deep Learning Homework"
```

```
In [ ]: !uv pip install -r requirements.txt --no-progress
```

Part1: Implementing CNN from Scratch

```
In [6]: PART_ONE_FULL_PATH = FULL_PATH + "/part1-convnet/"
```

```
In [ ]: %cd $PART_ONE_FULL_PATH
```

Load CIFAR-10 data

```
In [ ]: %cd data
!python get_data.py
%cd ..
```

Cross-check your implementation

```
In [ ]: # uncomment to run individual tests
!python -m pytest tests/test_conv.py -v
!python -m pytest tests/test_linear.py -v
!python -m pytest tests/test_maxpool.py -v
!python -m pytest tests/test_relu.py -v
!python -m pytest tests/test_sgd.py -v
```

```
In [ ]: # run all tests
!python -m pytest tests -v
```

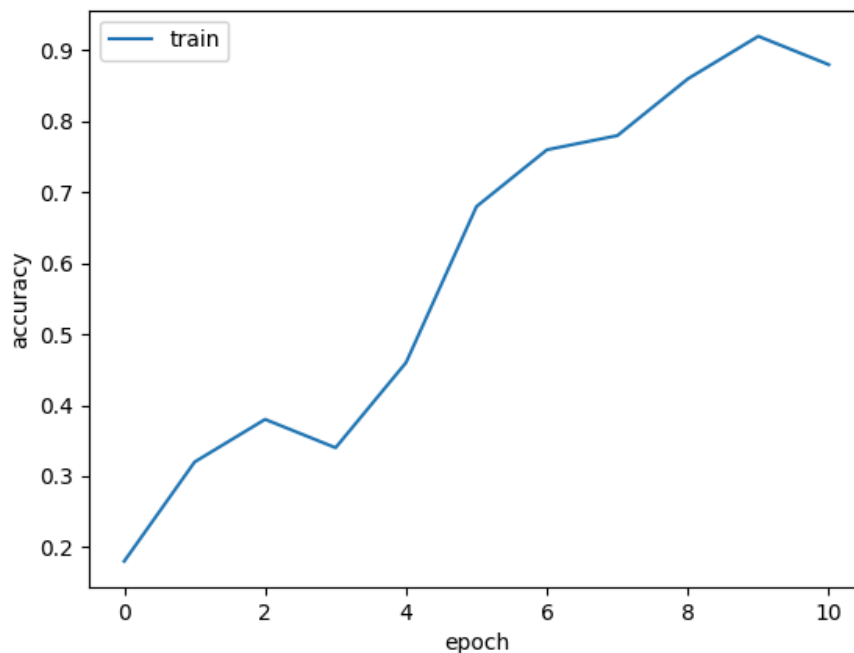
Train your ConvNet

```
In [ ]: !python train.py
```

```
In [18]: from utils import load_training_curve
```

Visualize Part-1 training curve

```
In [19]: load_training_curve("./train.png")
```



Part2: PyTorch

```
In [20]: PART_TWO_FULL_PATH = FULL_PATH + "/part2-pytorch/"
```

```
In [ ]: %cd $PART_TWO_FULL_PATH
```

Load CIFAR-10 data

```
In [ ]: %cd data
!python get_data.py
%cd ..
```

Train your Two-Layer Net

```
In [ ]: !python main.py --config configs/config_twolayer.yaml
```

Train your Vanilla ConvNet

```
In [ ]: !python main.py --config configs/config_vanilla_cnn.yaml
```

```
In [ ]: # uncomment to run individual tests
!python -m pytest tests/test_twolayer.py -v
!python -m pytest tests/test_vanilla_cnn.py -v
```

```
In [ ]: # run all tests
!python -m pytest tests -v
```

Train your own model

```
In [ ]: !python main.py --config configs/config_mymodel.yaml
```

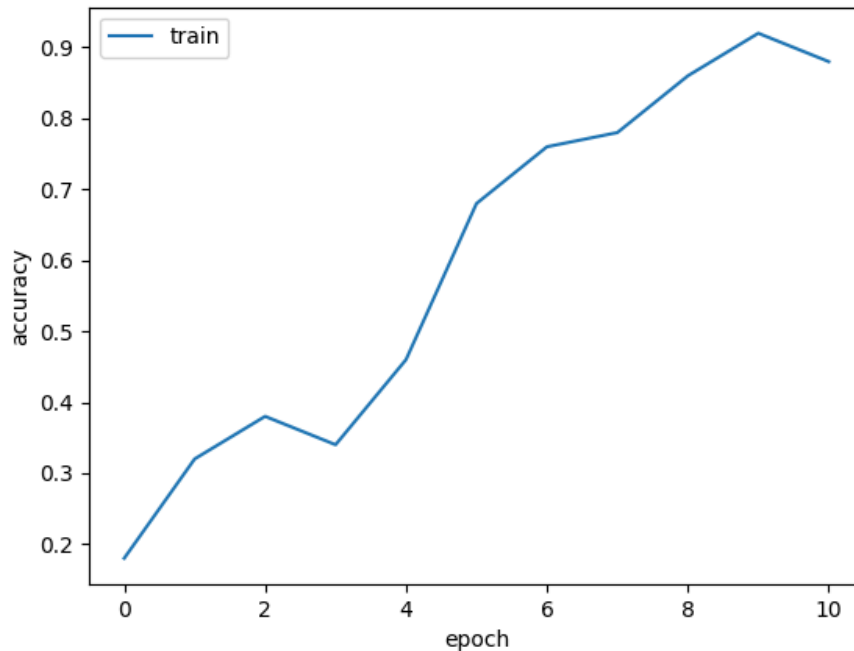
Assignment 2 Writeup

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Plot the training curve for Part-1.

```
In [40]: %cd $PART_ONE_FULL_PATH
from utils import load_training_curve
load_training_curve("./train.png")
```

/content/drive/MyDrive/Deep Learning Homework/hw2/part1-convnet



Briefly describe your design for the architecture of MyModel. What are the key components/characteristics that make it work?

MyModel, as I architected it, is a convolutional neural network designed with two sequential convolutional blocks followed by a fully connected classification head:

Dual-Stage Feature Extractor: The first block uses two 3×3 convolutional layers (32 channels), and the second block doubles the capacity to 64 channels across two more 3×3 layers. Small 3×3 filters maintain a large effective receptive field while keeping parameter count and compute low.

Batch Norm: Applied after every convolutional layer and before the final classification layer. BatchNorm stabilizes internal covariate shift, enabling faster convergence and allowing the use of higher learning rates, which I had to resort to after a few experiments, considering the pace of learning in SGD.

Spatial Downsampling: A 2×2 MaxPool layer (stride 2) at the end of each block reduces spatial dimensions ($32 \times 32 \rightarrow 16 \times 16 \rightarrow 8 \times 8$), condensing feature representations and providing translational invariance.

Regularization: Spatial Dropout2d (0.2 and 0.3) after pooling blocks and standard Dropout (0.5) in the dense layer prevent co-adaptation of features, mitigating overfitting on CIFAR-10.

When designing MyModel, did you try anything i.e. (making the model deeper, adding specific layers/activations, etc.) that didn't end up working in the way you expected? Why might that be?

Initially, using a very small learning rate ($lr=10^{-4}$) with default SGD momentum resulted in almost negligible progress across early epochs, as the gradient updates were too small to overcome the parameter volume introduced by multiple convolutional and batch norm layers. Increasing the learning rate to 0.05 with momentum (0.9) was necessary to escape flat loss regions.

Additionally, experimenting with class-balanced Focal Loss on standard training distributions did not yield expected gains compared to standard Cross-Entropy. My judgement is, because Focal Loss scales loss gradients by $(1-p_t)^\gamma$, it down-weighted well-classified or moderately confident examples.

On a balanced dataset with diverse classes, this significantly attenuated the overall gradient signal, leading to slower convergence compared to Cross-Entropy unless compensated for with extra efforts in loss scaling or extended training schedules.

Submission

IMPORTANT: Before submitting, please clear all output cells besides the plot for part 1 showing the training curve.

This cell will zip up your code for part 1 and part 2 into separate zip files as well as generate the notebook pdf.

Colab may take some time to update on the cloud file system, so if you find that running this cell does not generate the latest version of your notebook, you can wait a few minutes before trying again, or just download the notebook and run the command locally.

```
In [ ]: # on colab, you may need to install chromium to convert the notebook to pdf  
!playwright install-deps chromium  
!playwright install chromium
```

```
Installing dependencies...
Get:1 https://cli.github.com/packages stable InRelease [3,917 B]
Get:2 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64 InRelease
[1,578 B]
Get:3 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble InRelease [17.8 kB]
Hit:4 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu noble InRelease
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:6 http://archive.ubuntu.com/ubuntu noble InRelease
Get:7 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:8 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64 Packages
[1,877 kB]
Get:9 https://cloud.r-project.org/bin/linux/ubuntu noble-cran40/ InRelease [3,631 B]
Get:10 https://cli.github.com/packages stable/main amd64 Packages [359 B]
Get:11 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu noble/main amd64 Packages [52.
1 kB]
Get:12 https://r2u.stat.illinois.edu/ubuntu noble InRelease [9,161 B]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:14 https://cloud.r-project.org/bin/linux/ubuntu noble-cran40/ Packages [73.2 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,905 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [56.2 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [2,160 kB]
Get:18 https://r2u.stat.illinois.edu/ubuntu noble/main all Packages [10.2 MB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,299 kB]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,547 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,633 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2,048 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [49.0 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [36.0 kB]
Get:25 https://r2u.stat.illinois.edu/ubuntu noble/main amd64 Packages [3,023 kB]
Fetched 26.4 MB in 3s (9,434 kB/s)
Reading package lists... Done
W: https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2404/x86_64/InRelease: Key
is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section
in apt-key(8) for details.
W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.sta
t.illinois.edu/ubuntu noble InRelease' does not seem to provide it (sources.list entry missp
elt?)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libasound2t64 is already the newest version (1.2.11-1ubuntu0.3).
libasound2t64 set to manually installed.
libcairo2 is already the newest version (1.18.0-3build1).
libcairo2 set to manually installed.
libcups2t64 is already the newest version (2.4.7-1.2ubuntu7.14).
libcups2t64 set to manually installed.
libdbus-1-3 is already the newest version (1.14.10-4ubuntu4.1).
libdbus-1-3 set to manually installed.
libdrm2 is already the newest version (2.4.125-1ubuntu0.1~24.04.2).
libdrm2 set to manually installed.
libgbm1 is already the newest version (25.2.8-0ubuntu0.24.04.2).
libgbm1 set to manually installed.
libnspr4 is already the newest version (2:4.35-1.1build1).
libnspr4 set to manually installed.
libnss3 is already the newest version (2:3.98-1ubuntu0.2).
libnss3 set to manually installed.
```

libpango-1.0-0 is already the newest version (1.52.1+ds-1build1).
libpango-1.0-0 set to manually installed.
libx11-6 is already the newest version (2:1.8.7-1build1).
libx11-6 set to manually installed.
libxcb1 is already the newest version (1.15-1ubuntu2).
libxcb1 set to manually installed.
libxdamage1 is already the newest version (1:1.1.6-1build1).
libxdamage1 set to manually installed.
libxext6 is already the newest version (2:1.3.4-1build2).
libxfixed3 is already the newest version (1:6.0.0-2build1).
libxfixed3 set to manually installed.
libxkbcommon0 is already the newest version (1.6.0-1build1).
libxkbcommon0 set to manually installed.
libxrandr2 is already the newest version (2:1.5.2-2build1).
libxrandr2 set to manually installed.
libfontconfig1 is already the newest version (2.15.0-1ubuntu2).
libfreetype6 is already the newest version (2.13.2+dfsg-1ubuntu0.1).
libfreetype6 set to manually installed.
fonts-liberation is already the newest version (1:2.1.5-3).
The following additional packages will be installed:
 at-spi2-common gir1.2-glib-2.0 gir1.2-glib-2.0-dev libgirepository-2.0-0
 libglib2.0-bin libglib2.0-data libglib2.0-dev libglib2.0-dev-bin
 xfonts-encodings xfonts-utils xserver-common
Suggested packages:
 low-memory-monitor libglib2.0-doc libgdk-pixbuf2.0-bin libxml2-utils
Recommended packages:
 fonts-ipafont-mincho fonts-tlwg-loma at-spi2-core xdg-user-dirs xfonts-base
The following NEW packages will be installed:
 at-spi2-common fonts-freefont-ttf fonts-ipafont-gothic
 fonts-noto-color-emoji fonts-tlwg-loma-otf fonts-unifont fonts-wqy-zenhei
 libatk-bridge2.0-0t64 libatk1.0-0t64 libatspi2.0-0t64 libxcomposite1
 xfonts-cyrillic xfonts-encodings xfonts-scalable xfonts-utils
The following packages will be upgraded:
 gir1.2-glib-2.0 gir1.2-glib-2.0-dev libgirepository-2.0-0 libglib2.0-0t64
 libglib2.0-bin libglib2.0-data libglib2.0-dev libglib2.0-dev-bin
 xserver-common xvfb
10 upgraded, 15 newly installed, 0 to remove and 112 not upgraded.
Need to get 36.8 MB of archives.
After this operation, 91.2 MB of additional disk space will be used.
Get:1 <http://archive.ubuntu.com/ubuntu> noble/universe amd64 fonts-ipafont-gothic all 00303-2
1ubuntu1 [3,513 kB]
Get:2 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 gir1.2-glib-2.0-dev amd64 2.
80.0-6ubuntu3.9 [848 kB]
Get:3 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libglib2.0-data all 2.80.0-6
ubuntu3.9 [50.1 kB]
Get:4 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libglib2.0-dev amd64 2.80.0-
6ubuntu3.9 [1,860 kB]
Get:5 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libglib2.0-bin amd64 2.80.0-
6ubuntu3.9 [97.9 kB]
Get:6 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libgirepository-2.0-0 amd64
2.80.0-6ubuntu3.9 [73.5 kB]
Get:7 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libglib2.0-dev-bin amd64 2.8
0.0-6ubuntu3.9 [138 kB]
Get:8 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 gir1.2-glib-2.0 amd64 2.80.0
-6ubuntu3.9 [183 kB]
Get:9 <http://archive.ubuntu.com/ubuntu> noble-updates/main amd64 libglib2.0-0t64 amd64 2.80.0

```

-6ubuntu3.9 [1,547 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble/main amd64 at-spi2-common all 2.52.0-1build1
[8,674 B]
Get:11 http://archive.ubuntu.com/ubuntu noble/main amd64 fonts-freefont-ttf all 20211204+svn
4273-2 [5,641 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 fonts-noto-color-emoji all
2.047-0ubuntu0.24.04.1 [9,764 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble/universe amd64 fonts-tlwg-loma-otf all 1:0.7.3
-1 [107 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble/universe amd64 fonts-unifont all 1:15.1.01-1bu
ild1 [2,993 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble/universe amd64 fonts-wqy-zenhei all 0.9.45-8
[7,472 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble/main amd64 libatk1.0-0t64 amd64 2.52.0-1build1
[55.3 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble/main amd64 libatspi2.0-0t64 amd64 2.52.0-1buil
d1 [80.5 kB]
Get:18 http://archive.ubuntu.com/ubuntu noble/main amd64 libatk-bridge2.0-0t64 amd64 2.52.0-
1build1 [66.0 kB]
Get:19 http://archive.ubuntu.com/ubuntu noble/main amd64 libxcomposite1 amd64 1:0.4.5-1build
3 [6,320 B]
Get:20 http://archive.ubuntu.com/ubuntu noble/main amd64 xfonts-encodings all 1:1.0.5-0ubunt
u2 [578 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble/main amd64 xfonts-utils amd64 1:7.7+6build3 [9
4.4 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble/universe amd64 xfonts-cyrillic all 1:1.0.5+nmu
1 [384 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble/main amd64 xfonts-scalable all 1:1.0.3-1.3 [30
4 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 xserver-common all 2:21.1.1
2-1ubuntu1.8 [34.8 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 xvfb amd64 2:21.1.12-1u
buntu1.8 [877 kB]
Fetched 36.8 MB in 2s (21.1 MB/s)
Selecting previously unselected package fonts-ipafont-gothic.
(Reading database ... 126952 files and directories currently installed.)
Preparing to unpack .../00-fonts-ipafont-gothic_00303-21ubuntu1_all.deb ...
Unpacking fonts-ipafont-gothic (00303-21ubuntu1) ...
Preparing to unpack .../01-gir1.2-glib-2.0-dev_2.80.0-6ubuntu3.9_amd64.deb ...
Unpacking gir1.2-glib-2.0-dev:amd64 (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../02-libglib2.0-data_2.80.0-6ubuntu3.9_all.deb ...
Unpacking libglib2.0-data (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../03-libglib2.0-dev_2.80.0-6ubuntu3.9_amd64.deb ...
Unpacking libglib2.0-dev:amd64 (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../04-libglib2.0-bin_2.80.0-6ubuntu3.9_amd64.deb ...
Unpacking libglib2.0-bin (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../05-libgirepository-2.0-0_2.80.0-6ubuntu3.9_amd64.deb ...
Unpacking libgirepository-2.0-0:amd64 (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../06-libglib2.0-dev-bin_2.80.0-6ubuntu3.9_amd64.deb ...
/usr/bin/py3clean:101: DeprecationWarning: glob.glob1 is deprecated and will be removed in P
ython 3.15. Use glob.glob and pass a directory to its root_dir argument instead.
    for fn in glob1(directory, "%s.*" % fname):
Unpacking libglib2.0-dev-bin (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../07-gir1.2-glib-2.0_2.80.0-6ubuntu3.9_amd64.deb ...
Unpacking gir1.2-glib-2.0:amd64 (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Preparing to unpack .../08-libglib2.0-0t64_2.80.0-6ubuntu3.9_amd64.deb ...

```


Unpacking libglib2.0-0t64:amd64 (2.80.0-6ubuntu3.9) over (2.80.0-6ubuntu3.8) ...
Selecting previously unselected package at-spi2-common.
Preparing to unpack .../09-at-spi2-common_2.52.0-1build1_all.deb ...
Unpacking at-spi2-common (2.52.0-1build1) ...
Selecting previously unselected package fonts-freefont-ttf.
Preparing to unpack .../10-fonts-freefont-ttf_20211204+svn4273-2_all.deb ...
Unpacking fonts-freefont-ttf (20211204+svn4273-2) ...
Selecting previously unselected package fonts-noto-color-emoji.
Preparing to unpack .../11-fonts-noto-color-emoji_2.047-0ubuntu0.24.04.1_all.deb ...
Unpacking fonts-noto-color-emoji (2.047-0ubuntu0.24.04.1) ...
Selecting previously unselected package fonts-tlwg-loma-otf.
Preparing to unpack .../12-fonts-tlwg-loma-otf_1%3a0.7.3-1_all.deb ...
Unpacking fonts-tlwg-loma-otf (1:0.7.3-1) ...
Selecting previously unselected package fonts-unifont.
Preparing to unpack .../13-fonts-unifont_1%3a15.1.01-1build1_all.deb ...
Unpacking fonts-unifont (1:15.1.01-1build1) ...
Selecting previously unselected package fonts-wqy-zenhei.
Preparing to unpack .../14-fonts-wqy-zenhei_0.9.45-8_all.deb ...
Unpacking fonts-wqy-zenhei (0.9.45-8) ...
Selecting previously unselected package libatk1.0-0t64:amd64.
Preparing to unpack .../15-libatk1.0-0t64_2.52.0-1build1_amd64.deb ...
Unpacking libatk1.0-0t64:amd64 (2.52.0-1build1) ...
Selecting previously unselected package libatspi2.0-0t64:amd64.
Preparing to unpack .../16-libatspi2.0-0t64_2.52.0-1build1_amd64.deb ...
Unpacking libatspi2.0-0t64:amd64 (2.52.0-1build1) ...
Selecting previously unselected package libatk-bridge2.0-0t64:amd64.
Preparing to unpack .../17-libatk-bridge2.0-0t64_2.52.0-1build1_amd64.deb ...
Unpacking libatk-bridge2.0-0t64:amd64 (2.52.0-1build1) ...
Selecting previously unselected package libxcomposite1:amd64.
Preparing to unpack .../18-libxcomposite1_1%3a0.4.5-1build3_amd64.deb ...
Unpacking libxcomposite1:amd64 (1:0.4.5-1build3) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../19-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../20-xfonts-utils_1%3a7.7+6build3_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build3) ...
Selecting previously unselected package xfonts-cyrillic.
Preparing to unpack .../21-xfonts-cyrillic_1%3a1.0.5+nmu1_all.deb ...
Unpacking xfonts-cyrillic (1:1.0.5+nmu1) ...
Selecting previously unselected package xfonts-scalable.
Preparing to unpack .../22-xfonts-scalable_1%3a1.0.3-1.3_all.deb ...
Unpacking xfonts-scalable (1:1.0.3-1.3) ...
Preparing to unpack .../23-xserver-common_2%3a21.1.12-1ubuntu1.8_all.deb ...
Unpacking xserver-common (2:21.1.12-1ubuntu1.8) over (2:21.1.12-1ubuntu1.6) ...
Preparing to unpack .../24-xvfb_2%3a21.1.12-1ubuntu1.8_amd64.deb ...
Unpacking xvfb (2:21.1.12-1ubuntu1.8) over (2:21.1.12-1ubuntu1.6) ...
Setting up fonts-noto-color-emoji (2.047-0ubuntu0.24.04.1) ...
Setting up fonts-wqy-zenhei (0.9.45-8) ...
Setting up fonts-freefont-ttf (20211204+svn4273-2) ...
Setting up libglib2.0-0t64:amd64 (2.80.0-6ubuntu3.9) ...
Setting up libglib2.0-data (2.80.0-6ubuntu3.9) ...
Setting up fonts-tlwg-loma-otf (1:0.7.3-1) ...
Setting up libatspi2.0-0t64:amd64 (2.52.0-1build1) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up gir1.2-glib-2.0:amd64 (2.80.0-6ubuntu3.9) ...

```
Setting up fonts-ipafont-gothic (00303-21ubuntu1) ...
update-alternatives: using /usr/share/fonts/opentype/ipafont-gothic/ipag.ttf to provide /usr/share/fonts/truetype/fonts-japanese-gothic.ttf (fonts-japanese-gothic.ttf) in auto mode
Setting up gir1.2-glib-2.0-dev:amd64 (2.80.0-6ubuntu3.9) ...
Setting up at-spi2-common (2.52.0-1build1) ...
Setting up xserver-common (2:21.1.12-1ubuntu1.8) ...
Setting up libxcomposite1:amd64 (1:0.4.5-1build3) ...
Setting up fonts-unifont (1:15.1.01-1build1) ...
Setting up libglib2.0-dev-bin (2.80.0-6ubuntu3.9) ...
Setting up libatk1.0-0t64:amd64 (2.52.0-1build1) ...
Setting up libgirepository-2.0-0:amd64 (2.80.0-6ubuntu3.9) ...
Setting up xvfb (2:21.1.12-1ubuntu1.8) ...
Setting up libglib2.0-bin (2.80.0-6ubuntu3.9) ...
Setting up xfonts-utils (1:7.7+6build3) ...
Setting up libglib2.0-dev:amd64 (2.80.0-6ubuntu3.9) ...
Setting up xfonts-cyrillic (1:1.0.5+nmu1) ...
Setting up xfonts-scalable (1:1.0.3-1.3) ...
Setting up libatk-bridge2.0-0t64:amd64 (2.52.0-1build1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...
/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opengl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

Processing triggers for man-db (2.12.0-4build2) ...
```

```
In [ ]: %cd $FULL_PATH
        !python collect_submission.py
```